



EDU TERIA

72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE

19

**UNIT AND DIMENSION, PHOTOELECTRIC EFFECT
WPE, NUCLEAR FISSION & FUSION, MESUREMENT**



Unit and Dimension, Measurement

Unit and Measurements

- To measure physical quantity like length, mass and time, we require a standard of measurement.
- This standard of measurement is called the unit of that physical quantity. **For example**, the unit of length is metre, and a standard length of 1 metre has a precise definition.
- To measure the length of an object, we need to determine how many times this standard-length metre is contained in the length of the object.
- The comparison of a physical quantity with a standard quantity is called measurement.

Physical Quantities

- Those quantities which can describe the laws of physics are called the physical quantity. A physical quantity is one that can be measured. Thus, length, mass, time, pressure, temperature, current and resistance are considered as physical quantities.

Classification of Physical Quantities

The physical quantities are classified into :

- Fundamental quantities or base quantities:** The physical quantities that are independent of each other are called fundamental quantities.
- Derived quantities:** All the other quantities which can be expressed in terms of the fundamental quantities are called the derived quantities.

Units

- The reference standard used to measure the physical quantities is called the unit.

Properties of Unit

- The unit should be of some suitable size.
- The unit must be well-defined.
- The unit should be easily reproducible, i.e., it should not change with place.
- The unit must not change with time.
- The unit should not change with physical conditions like temperature, pressure, etc.

- The unit must be easily comparable experimentally with similar physical

Types of Units

(i) Fundamental Units

- The units defined for the fundamental quantities are called fundamental units.

(ii) Derived Units

- The units of all other physical quantities which are derived from the fundamental units are called the derived units.

System of Units

- FPS System :** In this system, the unit of length is foot, the unit of mass is pound, and the unit of time is second.
- CGS System :** In this system, the units of length, mass and time are centimeter, gram and second, respectively.
- MKS System:** In this system, the unit of length, mass and time are meter, kilogram and second, respectively.
- SI System:** This system is widely used in all measurements throughout the world. units. The system is based on seven basic units and two supplementary

SI Basic Units		
Quantity	Unit	Symbol of the Unit
Length	Metre	M
Mass	Kilogram	Kg
Time	Second	S
Temperature	Kelvin	K
Electric Current	Ampere	A
Amount of substances	Mole	Mol
Luminous Intensity	Candela	cd

Supplementary Units		
Quantity	Unit	Symbol of the unit
Plane angle	Radian	Rad
Solid angle	Steradian	Sr

Units Properties of Unit

- Metre (m) :** One metre is the distance travelled by light in the vacuum during a time interval of $(1/299792458)$ seconds.
- Kilogram (kg):** It is the mass of a platinum-iridium cylinder kept at the National Bureau of Weights and Measurements, Paris.
- Second (s):** The second is the time taken by the light of a specified wavelength emitted by a caesium-133 atom to execute 9192631770 vibrations.
- Ampere (A):** One ampere is that current which, when passed through two straight parallel conductors of infinite length and of negligible cross-section kept at 1A. meter apart in the vacuum, produces between them a force equal to 2×10^{-7} newton per metre length.
- Kelvin (K):** It is the fraction $1/273.6$ of the thermodynamic temperature of the triple point of water.
- Candela (cd):** A candela is defined as $1/60^{\text{th}}$ of the luminous intensity of 1 square centimetre of a perfect black body maintained at the freezing temperature of platinum (1773°C).
- Mole (md):** One mole is the amount of substance that contains elementary units equal to the number of atoms in 0.012 kg of carbon-12.

Supplementary Units

- Radian (rad):** The radian is the angle subtended at the centre of the circle by the arc whose length is equal to the radius of the circle.
- Steradian (Sr):** The steradian is the solid angle subtended at the centre of a sphere by a spherical surface of an area equal to the square of its radius.

Dimensional Formula

The dimensional formula of any physical quantity is the formula that tells which of the fundamental units have been used for the measurement of that physical quantity.

Q. How is dimensional formula written for a physical quantity?

- The formula of physical quantity must be written. The quantity must be on the left-hand side of the equation.
- All the quantities on the right-hand side of the formula must be written in terms of fundamental quantities like mass, length and time.

Physical Quantity	Unit	Dimension
Distance/Displacement	Meter-m	$M^0L^1T^0$
Speed/Velocity	M/s	$M^0L^1T^{-1}$
Acceleration (a)	$\Delta V \text{ m/s m}$	$M^0L^1T^{-2}$
Momentum (p)	Mass $\times$ velocity	$M^1L^1T^{-1}$
Force (f)	Mass $\times$ acceleration ($m \times a$) kg m/s ² —Newton	$M^1L^1T^{-2}$
Work (w)	$\vec{F} \cdot \vec{S}$ (Force Displacement) Kg m/s ² $\times$ m	$M^1L^2T^{-2}$
Gravitational constant (G)	$G = \frac{F \times m^2}{kg^2}$ $\frac{kgm/s^2 \times m^2}{kg^2} = \frac{m^3}{kg s^2}$	$M^{-1}L^3T^{-2}$
Electrostatic constant (k)	$F = \frac{kq_1q_2}{r^2}$ $K = \frac{Fr^2}{q_1q_2}$ $\Rightarrow \frac{kgm/s^2 \times m^2}{c^2 \times s^2 \times A^2} = \frac{kgm^3}{s^4 \times A^2}$ $q=IT$ $C=As$	$M^1L^3T^{-4}A^{-2}$
Torque (Similar to work)	$\vec{r} \times \vec{F}$ Length force $m \times \frac{kgm}{s^2}$	$M^1L^2T^{-2}$
Impulse (I) (change in maximum)	Δp —kgm/s	$M^1L^1T^{-1}$
Strain	$\frac{\text{change in length}}{\text{original length}} = \frac{\Delta l}{l} = \frac{m}{m}$	Unitless Dimension less

The dimensional formula of any physical quantity is the formula that tells which of the fundamental units have been used for the measurement of that physical quantity.

Q. How is dimensional formula written for a physical quantity?



- (3) Replace mass, length and time with M, L and T, respectively.
- (4) Write the powers of the terms.

Characteristics of Dimensions :—

- Dimensions do not depend on the system of units.
- Quantities with similar dimensions can be added or subtracted from each other.
- Dimensions can be obtained from the units of the physical quantities and vice versa
- Two different quantities can have the same dimension.
- When two dimensions are multiplied or divided, it will form the dimension of the third quantity.

The dimensional formula can be used to

- Check the correctness of the equation.
- Convert the unit of the physical quantity from one system to another.
- Deduce the relation connecting the physical quantities.

Units and Dimensions of a Few Derived Quantities		
Physical Quantity	Unit	Dimensional Formula
Displacement	m	$M^0 L^1 T^0$
Area	m^2	$M^0 L^2 T^0$
Volume	m^3	$M^0 L^3 T^0$
Velocity	ms^{-1}	$M^0 L^1 T^{-1}$
Acceleration	ms^{-2}	$M^0 L^1 T^{-2}$
Density	kgm^{-3}	$M^1 L^{-3} T^0$
Momentum	$kg ms^{-1}$	$M^1 L^1 T^{-1}$
Work/Energy/Heat	Joule (or) $kg m^2/sec^2$	$M^1 L^2 T^{-2}$
Power	Watt(w)(or) $Joule/sec$	$M^1 L^2 T^{-3}$
Angular Velocity	$rad s^{-1}$	$M^0 L^0 T^{-1}$
Angular Acceleration	$rad s^{-2}$	$M^0 L^0 T^{-2}$
Resistance/impedance	Ohm(or) $kgm^2/sec C^2$	$ML^2T^{-1}Q^{-2}$
EMF/voltage/potential	Volt (or) $kg m^2/sec^2C$	$ML^2T^{-2}Q^{-1}$
Permeability	Henry/m (or) $kg m/C^2$	MLQ^{-2}
Permittivity	Farad/m (or) sec^2C^2/kgm^3	$T^2Q^2M^{-1}L^{-3}$
Frequency	Hertz (or) sec^{-1}	T^{-1}
Wavelength	m	L^1
Moment of inertia	$Kg m^2$	$M^1 L^2 T^0$

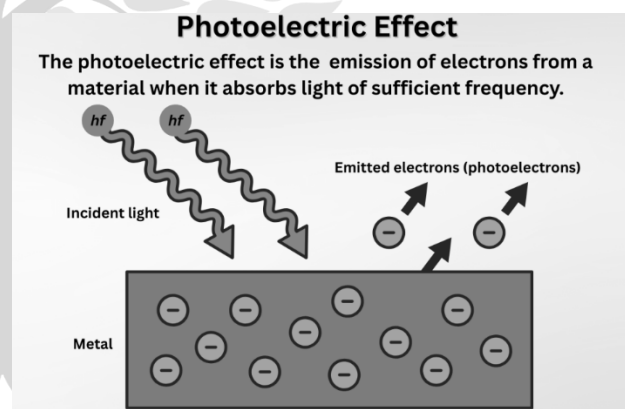
Force	Newton (or) kgm/sec^2	$M^1 L^1 T^{-2}$
Pressure	Newton/m(or) $Kg m^{-1}/sec^2$	$M^1 L^{-1} T^{-2}$
Impulse	Newton sec (or) $kg m/sec$	$M^1 L^1 T^{-1}$
Inertia	$Kg m^2$	$M^1 L^2 T^0$
Electric current	Ampere (or) C/sec	QT^{-1}

PHOTOELECTRIC EFFECT

Emission of electrons from a metal surface when electromagnetic radiation of sufficient frequency falls on it.

Experimental Observations

- Threshold Frequency (ν_0):** Below ν , no photoemission occurs, regardless of intensity.
- Instantaneous emission:** No time lag between light incidence and emission.
- Kinetic energy depends on frequency, not intensity.
- Photoelectric current depends on intensity, not frequency.



Einstein's Photoelectric Equation

$$h\nu = \phi + K_{\max}$$

Where:

- $h\nu$ = energy of incident photon
- $\phi = h\nu_0$ = work function of metal
- $K_{\max} = \frac{1}{2}mv_{\max}^2 = eV_0$

Significance:

- Confirms particle nature of light
- Photon has energy $E = h\nu$

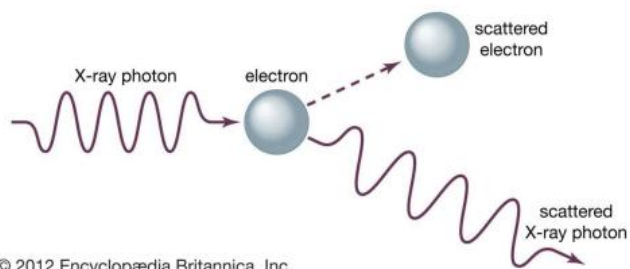
Photon

- Quantum (packet) of electromagnetic radiation
- Properties:





- Energy = $h\nu$
- Momentum = $p = h/\lambda$
- Zero rest mass
- Travels at speed of light



WAVE-PARTICLE DUALITY

Light behaves as:

- Wave interference, diffraction
- Particle photoelectric, Compton effect

DE BROGLIE HYPOTHESIS (MATTER WAVES)

Every moving particle is associated with a wave.

De Broglie Wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

For electron accelerated through potential V:

$$\lambda = \frac{h}{\sqrt{2meV}}$$

ATOMIC NUCLEUS

Radioactive Decay Law

$$N = N_0 e^{-\lambda t}$$

Half-life

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

Mean Life

$$\tau = \frac{1}{\lambda}$$

NUCLEAR FISSION & FUSION

Nuclear Fission

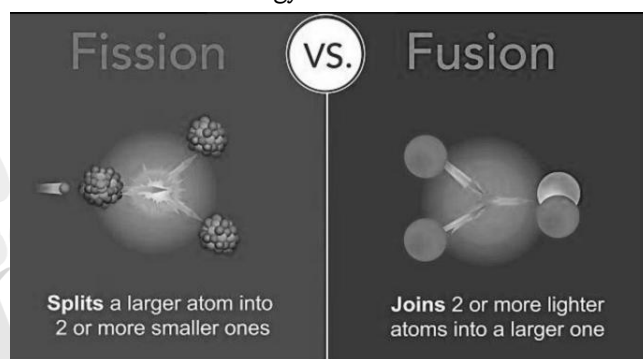
Nuclear fission is the process in which a heavy nucleus splits into two or more lighter nuclei, releasing a large amount of energy along with neutrons.

- Splitting of heavy nucleus (U-235)
- Used in nuclear reactors

Nuclear Fusion

Nuclear fusion is the process in which two light nuclei combine to form a heavier nucleus, releasing enormous energy.

- Combining light nuclei
- Source of energy in stars



Nuclear reactions				
Nuclear Fission		Nuclear Fusion		
neutron	U-235	deuterium	tritium	
U-236				
Ba-144	Neutrons			
Kr-89				
		neutron	helium	
Type	Nuclear equation	Representation	Change in mass/atomic numbers	
${}^4_2\text{He}$ Alpha decay	${}^A_Z\text{X} \rightarrow {}^{A-4}_{Z-2}\text{Y} + {}^4_2\text{He}$		A: decrease by 4 Z: decrease by 2	
${}^0_{-1}\text{e}$ Beta decay	${}^A_Z\text{X} \rightarrow {}^A_{Z+1}\text{Y} + {}^0_{-1}\text{e}$		A: unchanged Z: increase by 1	
Gamma decay	${}^A_Z\text{X} \rightarrow {}^A_Z\text{Y} + \gamma$		A: unchanged Z: unchanged	
${}^0_{+1}\text{e}$ Positron emission	${}^A_Z\text{X} \rightarrow {}^A_{Z-1}\text{Y} + {}^0_{+1}\text{e}$		A: unchanged Z: decrease by 1	
${}^0_0\gamma$ Electron capture	${}^A_Z\text{X} + {}^0_{-1}\text{e} \rightarrow {}^A_{Z-1}\text{Y} + \gamma$		A: unchanged Z: decrease by 1	

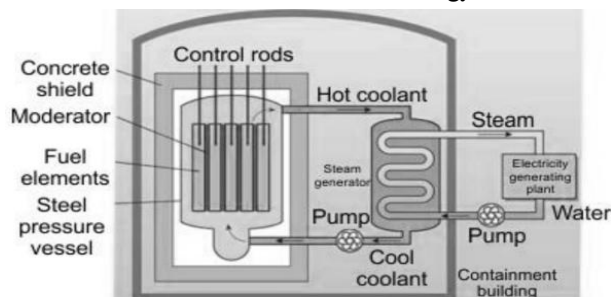
NUCLEAR REACTOR

- **Reactor Core:** Contains fuel rods (Uranium pellets) where fission occurs.
- **Fuel:** Enriched Uranium (U-235), Thorium (can be converted).
- **Moderator:** Slows down fast neutrons for sustained reaction (Heavy Water, Graphite, Light Water). Control
- **Control Rods:** Absorb neutrons to control reaction rate (Boron, Cadmium).
- **Coolant:** Removes heat (Water, Heavy Water). Steam



- **Generator:** Heat exchanger. Turbine & Generator:

Convert mechanical to electrical energy.



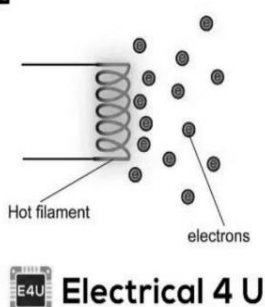
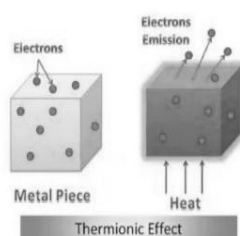
Thermionic Emission

Thermionic emission is the process by which electrons are emitted from the surface of a metal when it is heated sufficiently.

Basic Principle

- Metals contain free electrons.
- When heated, these electrons gain energy.
- If their energy exceeds the Work Function (minimum energy required to escape), they leave the metal surface.

What is Thermionic Emission?



Work, Power, and Energy

Work

Definition: Work is done when a force applied to a body causes a displacement in the direction of the force.

Formula: $W = \vec{F} \cdot \vec{S} = F.S \cos \theta$

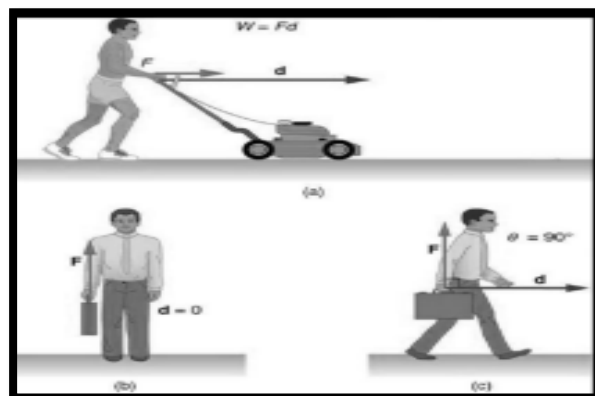
Unit: SI unit is the Joule (J); CGS unit is the erg (1 joule = 10^7 erg).

Quantity: It is a scalar quantity and can be positive, negative or zero. $\cos 180^\circ -1$

$W = \vec{F} \cdot \vec{S}$

SI Unit : Joule

Work Power and Energy



Energy

- Definition: Energy is the capacity to do work.
- Forms: Exists as kinetic (motion) and potential (position) energy, and many other forms (heat, light, chemical, etc.).
- Formula (Kinetic Energy): $KE = (1/2) mv^2$.

$$KE = \frac{1}{2}mv^2$$

- **Formula (Potential Energy):** $PE = mgh$.
- **Units:** SI unit is the Joule (J), the same as work.
- **Law of Conservation of Energy:** Energy cannot be created or destroyed, only transformed from one form to another.

Power

- **Definition:** Power is the rate of doing work or transferring energy.
- **Formula:** $P = W/t$ or $P = F \times v$ (force times velocity).
- **Units:** SI unit is the Watt (W), (1 Watt = 1 Joule per second); another common unit is horsepower (1 hp ~ 746 Watts).
- **Quantity:** It is a scalar quantity.



Device and their measurement

Device	Measure
Microscope	Magnification
Telescope	Distance objects
Spectrometer	Wavelength
Calorimeter	Heat energy
Thermometer	Temperature
Pyrometer	High temperature
Thermocouple	Temperature difference
Hygrometer	Humidity
Anemometer	Wind speed
Wind Vane	Wind direction
Rain Gauge	Rainfall
Lux Meter	Light intensity
Sound level meter	Sound
Barometer	Atmospheric pressure
Manometer	Gas pressure
Aneroid Barometer	Pressure
Speedometer	Speed
Odometer	Distance
Tachometer	Rotational speed
Accelerometer	Acceleration
Hydrometer	Density of liquid
Lactometer	Density of milk
Ammeter	Electric Current
Voltmeter	Voltage
Wattmeter	Electric Power
Energy Meter	Electrical Energy
Galvanometer	Small current
Ohmmeter	Resistance
Multimeter	Voltage, current & resistance
Potentiometer	Potential difference
Clamp Meter	Current (no contact)
Oscilloscope (CRO)	Voltage signal
Frequency Meter	Frequency
Megger	Insulation resistance